

Soheil Sayah Varg

✉ soheilsayahvarg@gmail.com • 🌐 github.com/soheilsayahvarg • 📞 +98 935 729 2456 • 🔗 linkedin.com/in/soheil-sayah

Education

Sharif University of Technology
B.Sc. in Computer Engineering

Tehran, Iran
Sep 2023 – Jul 2027

- **Cumulative GPA: 19.38 / 20.0**, ranked 15th of 183 students. Departmental average for this cohort: **17.24**; university-wide: 15.85.
- Graduate coursework taken as an undergraduate: **Stochastic Processes** (20.0), **Reinforcement Learning**.
- Awarded 7 credits in physics on admission under the National Olympiad medalist provision.

AE High School
Diploma, Mathematics & Physics

Tehran, Iran
Sep 2020 – Jun 2023

- GPA: 19.25 / 20.0

Honors and Awards

2023: Gold Medal, Iranian National Physics Olympiad — ranked 7th nationwide among more than 10,000 participants, and 2nd nationwide in the experimental (laboratory) examination.

2023: Admitted to Sharif University of Technology through the olympiad medalist track (no entrance exam).

Research Interests

Primary: Inference and approximation under uncertainty — when the object of interest cannot be computed exactly, what licenses trusting the approximation. Concretely: exact and approximate sampling for language models, and estimation for offline decision-making under partial observability.

Foundations: Statistical learning theory, reinforcement learning theory, and convex optimization — the tools these questions get asked in, and the source of the gap between what an estimator provably does and what it does at finite sample size.

Research Experience

Digital Media Lab (DML), Sharif University of Technology

Exact Controlled Generation from Tilted Autoregressive Targets

Dec 2025 – Present

- Supervised by **Dr. Ali Rostami**, in the research group of **Prof. Hamid R. Rabiee**.
- Sampling scheme that draws from a reward-tilted language-model distribution *exactly in distribution* rather than approximately, via a particle-Gibbs ladder over a batched proposal chain — so that steering a model's output does not silently change which distribution is being sampled from.
- Replaced the design's assumed theoretical scaling rate with a measured one across sequence lengths spanning a $32\times$ range, which moved the method's usable operating regime and re-anchored the design on measured rather than extrapolated points.
- Manuscript in preparation.

Graduate Deep RL project, team of 3, Sharif University of Technology

Model-Based Reinforcement Learning in Confounded POMDPs

Feb 2026 – Present

- Built the first empirical implementation of Hong, Qi & Xu (ICML 2024), a theory-only paper with no released code or experiments: dual bridge-function identification (reward and dynamics) from pre-decision negative-control proxies, oracle-verified to machine precision ($\sim 1e-15$).
- Found that the paper's pessimism step, implemented literally, *diverges* numerically, and built the repair the team settled on — **Signal-Projected Pessimism**, an SVD signal-subspace projection that restores zero regret on the tabular benchmark. Reported as a disclosed heuristic rather than a certified bound, since the true bridge leaks $\sim 21\%$ of its norm outside the projected subspace.
- Ran a 5-seed sweep at full benchmark scale (5 confounding levels $\times$ 3 sample sizes) and found the **confounding-blind baseline wins in 15/15 cells**: estimation variance dominates the bias the method is designed to remove. Reported as the headline result rather than the favorable subset.
- Extended the estimator to continuous state and action via an RKHS/kernel backend with a closed-form linear-Gaussian oracle; plug-in de-biasing beats naive OLS on 4/4 candidate policies with $3\text{--}11\times$ smaller bias.
- *My role in the team of three:* I read the two anchor papers and drove the implementation; the remaining literature and the correctness review of every estimator were shared with my teammates.

Selected Projects

Information Retrieval: Modern Information Retrieval System — Python, PyTorch, FAISS

🌐 repo

- Three-stage retrieval pipeline: SPLADE learned-sparse and dense retrieval fused by reciprocal rank fusion, then cross-encoder reranking; QLoRA fine-tuning for the generation stage. Evaluated with NDCG and MAP against a documented weak-supervision protocol.

Optimization: Convex Optimization Algorithms — Python, NumPy, CVXPY

🌐 repo

- First-order methods implemented from their update rules, with a solver used only as an oracle to disagree with. Chambolle-Pock primal-dual for TV denoising (adjoint identity verified to $1e-8$, step sizes taken from $\tau\sigma\|K\|^2 < 1$ rather than tuned) and ADMM for TV inpainting, which matches CVXPY to 0.02 dB PSNR while running in 0.2 s.

Applied Probability: Market Predictability Study — Python, statsmodels

 repo

- Independent re-examination, carried out after the course had ended, of my own profitable trading backtest — variance-ratio, Hurst, Ornstein-Uhlenbeck MLE, Newey-West with heteroskedasticity-consistent errors. Showed the claimed 5-tick lead signal was a multiple-comparisons artifact (scanning 21 lags of pure noise clears a single-test threshold 69.3% of the time) and that random entry reproduces the strategy's Sharpe ratio. The effect that survives is parameter staleness under drift: trailing re-estimation halves maximum drawdown.

Deep Learning: Medical Image Segmentation and Deep RL — PyTorch

 repo

- U-Net, Attention U-Net and Residual-Attention U-Net trained and compared for segmentation; Soft Actor-Critic implemented from scratch (twin critics, reparameterized policy, Polyak-averaged targets) for continuous control.

Systems: NeoGit — a version control system in C

 repo

- Git-like CLI built on raw POSIX file I/O: staging, commits, branching, merge, tagging, stashing, and pre-commit hooks, with an on-disk object layout of its own design.

Teaching Experience

2024 – 2026: Teaching Assistant, Sharif University of Technology — **nine course-semesters across five terms**, including **Head TA** for Signals & Systems (Dr. Manzuri, Spring 2026), coordinating the teaching staff for a 60-student core course.

Stochastic Processes and Engineering Probability & Statistics (Dr. Najafi) • Machine Learning (Dr. Sharifi-Zarchi) • Data Structures & Algorithms (Dr. Abam) • Linear Algebra (Dr. Rabiee & Dr. Ramezani) • Fundamentals of Programming (Dr. Rivadeh).

Aug 2023 – Present: Co-founder and physics instructor, Resonance (resonanceoly.ir) — an online Physics Olympiad school, started as a small study group, reaching students who have no selective high school near them. Teach the mechanics, electromagnetics and laboratory tracks, and write problem sets and mock exams for the national selection rounds.

2022 – Present: Physics Olympiad instructor, theory and **experimental / laboratory** tracks — AE High School and Mahdavi School; introductory Python at AE Middle School.

Academic Service and Outreach

May 2025: Juror, 13th Persian Young Naturalists' Tournament (PYNT), Tehran — an IYPT-format research tournament run by the Ariaian Young Innovative Minds Institute under the World Federation of Physics Competitions. Judged team defenses of open-ended physics research problems.

2024 – 2025: Deputy Chairman, CEI 2024 (Computer Engineering Introduction event) • **Head of Purchase**, WSS 11 (Winter Seminar Series) • Sponsorship Staff, WSS 10.

2023 – 2025: Head of Purchase & Scientific Staff, Bugsbuzzy game development contest • Executive Staff, ICPC 2023 & 2025 • Content & Social Support, Rayan AI 2024.

Selected Coursework (grades out of 20.0; [†] graduate-level)

- | | |
|---|---|
| ○ Stochastic Processes [†] : 20.0 | ○ Linear Algebra: 20.0 |
| ○ Machine Learning: 20.0 | ○ Probability & Statistics: 20.0 |
| ○ Artificial Intelligence: 20.0 | ○ Data Structures & Algorithms: 20.0 |
| ○ Convex Optimization I: 19.5 | ○ Signals & Systems: 20.0 |

Spring 2026: Reinforcement Learning[†], Modern Information Retrieval, Game Theory, Design of Algorithms, Database Design (*grades pending*).

Technical Skills

Languages: Python (proficient), C, C++, Java, SQL, MATLAB, $\LaTeX$

ML & Numerics: PyTorch, NumPy, SciPy, scikit-learn, statsmodels, pandas, Matplotlib, Hugging Face Transformers

Optimization & IR: CVXPY, FAISS, Gymnasium

Tools: Git, Linux, Docker, PostgreSQL, shell scripting, Proteus